

PROJECT SYNOPSIS · V0.2

Personal Agentic RAG

Fast and slow thinking for your studies and documents: private, measurable and free.

“Handle your documents safely and securely.”

Mohit & Kunal · Professor: Dr. Neetu Verma · DCRUST, Murthal



THE PROBLEM

Students' knowledge is scattered, and AI doesn't help safely



Scattered material

Notes, books, slides and past papers are spread across folders and phones



Hard to find

Marksheets, certificates and exam dates when you need them



Ungrounded answers

Chatbots answer from the internet, not your syllabus, and don't cite the page



Privacy cost

Cloud AI only works if your documents leave your control

Goal: an assistant that is **grounded**, **private** and **measurable**.

THE CORE IDEA

Think fast when you can, slow when you must

System 1 • Fast

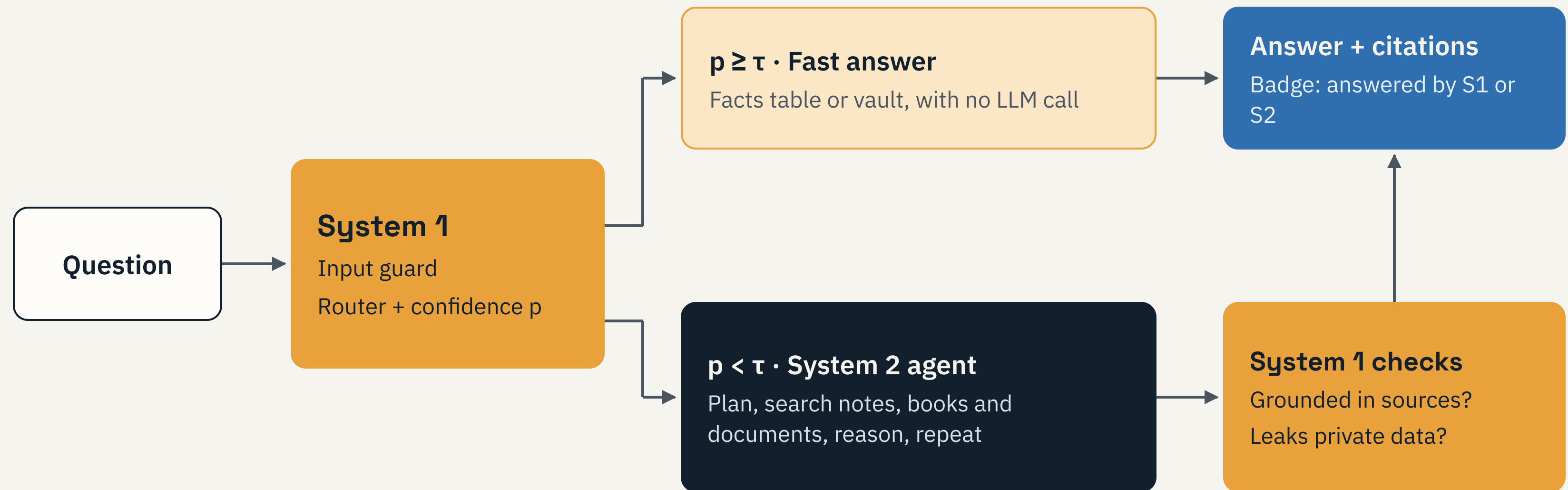
- Small model with typed answers + confidence
- Routes questions, answers simple facts
- Safety checks on every input and output
- No LLM call

System 2 • Slow

- LLM agent that plans and reasons
- Searches notes, books and documents
- Several steps, cites its sources
- Used only when System 1 is unsure

Inspired by Daniel Kahneman, *Thinking, Fast and Slow* (2011)

Escalate only when unsure



TWO EXAMPLES

Same assistant, two ways of thinking

FAST PATH · SYSTEM 1

“When is my DBMS exam?”

1. Guard: safe
2. Router: fact lookup, high confidence
3. Answer from the facts table
4. Cites the exam timetable PDF

No LLM call

SLOW PATH · SYSTEM 2

“Compare how my notes and Galvin explain paging”

1. Router: comparison, so escalate
2. Agent searches notes, then the book
3. Grounding check passes
4. Cites notes p. 14 and Galvin ch. 9

WHAT IT KNOWS

Your studies and your documents



Study library

- Lecture notes, books, slides
- Previous year question papers
- Organised: Semester → Course → Topic
- Answers cite the page number



Personal vault

- Marksheets, certificates, IDs
- Exam timetable and deadlines
- Encrypted, never sent to the cloud
- Facts extracted for fast answers

Scans and phone photos become searchable with **OCR**. Private data is detected automatically with **Presidio**.

GUARDRAILS AND SECURITY

Three System-1 guards, plus a locked vault

G1 · INPUT

Injection guard

Blocks attacks in questions and in text hidden inside uploaded PDFs

G2 · OUTPUT

Grounding check

Is every claim supported by a cited source?
If not, retry or say “I don’t know”

G3 · OUTPUT

PII leak check

Never reveals ID numbers or marks you didn’t ask for



AES-256 encrypted vault



Local models by default



Audit log + red-team tests

Every decision is traced

Example trace (illustrative)

```
▼ question
├ s1.guard.input → safe
├ s1.router → compare (low p)
├ s2.agent.plan
│ └ tool.library_search
│   └ tool.library_search
├ s1.guard.grounding → pass
└ s1.guard.pii → pass
```

Arize Phoenix

Open-source tracing that runs on the laptop, built on OpenTelemetry

Linked from the chat

Each answer shows its trace, path (S1/S2) and confidence

Private by default

PII is redacted from trace data

RESEARCH QUESTIONS

What the report will answer

ID	Question	Measured by
RQ1	Can System 1 answer some questions alone, as accurately as System 2?	% answered by S1, accuracy
RQ2	How much time and compute does escalation save?	Latency, LLM calls, tokens
RQ3	Does the agent beat plain RAG on our questions?	Correctness, faithfulness, page citations
RQ4	What do the guardrails catch, and what do they wrongly block?	Attack success, PII leaks, false refusals
RQ5	Which System-1 engine is better: jeff or our trained classifier?	Accuracy, calibration, speed

EVALUATION

Public data, clear baselines

Datasets

- “Alex Demo”: a fictional student’s documents
- Open textbooks (OpenStax)
- ~200 Q&A with gold pages
- ~100 red-team attacks

Compared systems

- Plain RAG
- System 2 only (agent)
- **System 1 + System 2 (ours)**
- Ours without guardrails
- jeff vs SetFit engine

Metrics

- Correctness, faithfulness (Ragas)
- Page-citation accuracy
- Latency, LLM calls
- Calibration (ECE)
- Attack success, false refusals

One command (**make eval**) reproduces every table and plot. No real personal data is used.

FREE FOR EVERY STUDENT

Open source, runs on an 8 GB laptop

O

cost. No GPU, no paid APIs.
An optional free cloud API
can be used for non-
sensitive data.

Part	Open-source tool
Chat UI	Chainlit
System 2 agent	LangGraph + Ollama (local LLM)
System 1	jeff · SetFit
Documents + OCR	Docling · Presidio
Storage	SQLite + LanceDB
Tracing	Arize Phoenix
Evals	Ragas + our harness

Two layers, one semester

MOHIT · KNOWLEDGE + SYSTEM 2

Ingestion and OCR · study library · encrypted vault and facts table · System-2 agent · Chainlit UI

KUNAL · SYSTEM 1 + TRUST

System-1 engines · escalation policy · guardrails · Phoenix tracing · evals and red-team

Week 4

Plain RAG baseline + first numbers

Week 6

System-1 fast answers working

Week 8

Mid-term demo: agent + guardrails

Week 12

Full results + report draft

Week 14

Final demo + viva

Where it can grow next



Study tools

Flashcards, practice quizzes from past papers, a study planner tied to exam dates



Career

Job-match score, tailored resume, LinkedIn drafts. You approve every action



Anywhere

Private cloud hosting with passkey login, and a mobile app

Thank you

Questions and feedback welcome

Open source from day one: github.com/MohitBareja16/personal-agentic-rag